

Optimizing Multi-Layer Radiation Absorbing Materials: Advanced Algorithmic Solutions

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Hello, we are Team 1. This semester, we had the privilege of collaborating with the Army Research Lab (DEVCOM) on an exciting project focused on designing an optimized Multilayered Radiation-Absorbing Material (RAM). The goal of this project was to minimize total electromagnetic wave reflection while ensuring the material thickness remained under 7 mm.

To achieve this, we employed advanced computational tools and techniques, developing two independent optimization frameworks in MATLAB. The first was a Genetic Algorithm, designed to simulate evolutionary principles and identify high-performing solutions through iterative refinement. The second was a Bayesian Optimization framework, a probabilistic model that efficiently searches for the global optimum. By implementing and analyzing these two approaches, we were able to compare their effectiveness, refine our designs, and identify the optimal method for enhancing the RAM's performance.

This project allowed us to apply engineering principles, computational methods, and innovative problem-solving strategies to a real-world challenge, showcasing the potential of advanced algorithms in material science applications.